DATABASE PROGRAMMING IN PYTHON

→python supports various databases like MYSQL, ORACLE, SYBASE, POSTGRESQL, etc……..

→python also supports DDL,DML and data query statements.

→python DB API is a widely used module that provides a database application programming interface.

Benefits:

1.python is famous for its portability.

2.it is platform independent.

3.python supports sql cursors.

4.python supports relational database management system.

5.python database API are compatable with various databases, so it is very easy to migrate and port databases applications interface.

Packages:

1.import mysql.connector:

This module API to connect MYSQL

2.from mysql.connector import error:

it shows us an error when we failed to connect database (or) if any other databases object error occurred while working with the databases.

Example:

ER_Access_DENIED_Error when username or password is wrong.

Arguments required to connect the MYSQL database from python:

(a)username:

the default username for the MYSQL database is 'root'.

(b)password:

it is given by the user at the time of installing the MYSQL database. if you are using root then you won't need the password.

(c)hostname:

it is server name (or) IP address on which MYSQL is running. If you are running on localhost, then you can use localhost (or) it's IP i.e 127.0.0.0.

(d)database name:
database name to which you want to connect here we are using database named "python_DB".

FUNCTIONS:
(1)mysql.connector.connect:
→this function we can connect the MYSQL database, this function accepts four required parameters:HOST,DATABASE, USERNAME, and PASSWORD that we already discussed.
→connect() function established a connection to the MYSQL database from python application and returned a MYSQLConnection object.
→connect() function throw an exception , i.e database error if one of the required parameter are wrong.
(2)conn.is_connected():
it is the function of the MYSQLCOnnection class through which we can verify is our python application connected to MYSQL.

(3)connection.cursor():
it returns a cursor object , using this object we can execute SQL queries. cursor objects interact with the MYSQL server using a MYSQLConnection object.

(4)cursor.close():
using this method we can close the cursor object. Once we close the cursor object , we cannot execute any SQL statement.

(5)connection.close():
closing the MYSQL databases connection using a close() function of MYSQLConnection class.

steps to connect MYSQL database in python using MYSQL connector PYTHON:
1.install MYSQL connector python using pip.
python –m pip install mysql-connector

2.use the mysql.connector.connect() method of MYSQL connector python with required parameters to connect MYSQL.
3.use the connection object returned by the connect() method to create a cursor object to perform database operations.
4.the cursor.execute() to execute SQL queries from python.
5.close the cursor object using a cursor.close() and MYSQL database connection using connection.close() after your work completes.
6.catch exception if any that may occur during this process.

program-1: (creating a database)

```
import mysql.connector
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar"
   )
mycursor=mydb.cursor()
mycursor.execute("create database hospital")
print("database created success.........!")
```

program-2: (creating a table in the hospital database)

```
import mysql.connector
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar",
   database="hospital"
   )
mycursor=mydb.cursor()
mycursor.execute("create table patients(name varchar(10),address
varchar(10))")
print("table created success.........!")
```

got mysql and type as follows:

```
mysql> use hospital;
Database changed
mysql> desc patients;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| name    | varchar(10) | YES  |     | NULL    |       |
| address | varchar(10) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
2 rows in set (0.16 sec)
```

Example-2: (inserting values into the table)

```
import mysql.connector
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar",
   database="hospital"
   )
mycursor=mydb.cursor()
sql="insert into patients(name,address)values(%s,%s)"
val=('john','vizag')
mycursor.execute(sql,val)
mydb.commit()
print(mycursor.rowcount,"record inserted")
```

goto mysql :

```
mysql> select * from patients;
+------+---------+
| name | address |
+------+---------+
| john | vizag   |
```

```
+------+---------+
1 row in set (0.00 sec)
```

Example-4:

```
import mysql.connector
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar",
   database="hospital"
   )
print("enter patient name:")
name=input()
print("enter patient address:")
address=input()

mycursor=mydb.cursor()
sql="insert into patients(name,address)values (%s,%s)"
val=(name,address)
mycursor.execute(sql,val)
mydb.commit()
print(mycursor.rowcount,"record inserted")
```

Example-5:

```
import mysql.connector
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar",
   database="hospital"
   )
```

```
mycursor=mydb.cursor()
sql="insert into patients(name,address)values (%s,%s)"
val=[('peter','usa'),
    ('amy','mumbai'),
    ('micheal','punjab'),
    ('sandy','assam'),
    ('vicky','delhi'),
    ('chuck','ooty')
    ]
mycursor.executemany(sql,val)
mydb.commit()
print("records inserted")
```

goto mysql:

```
select * from patients;
```

Example-6: (retrieving the data from backend to frondend)

```
import mysql.connector
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar",
   database="hospital"
   )

mycursor=mydb.cursor()
mycursor.execute("select * from patients")
myresult=mycursor.fetchall()
for x in myresult:
   print(x)
```

program-7: (updating data into the table)

```
import mysql.connector
```

```
mydb=mysql.connector.connect(
   host="localhost",
   user="root",
   passwd="manikumar",
   database="hospital"
   )

print("enter name:")
name=input()
print("enter address:")
address=input()

mycursor=mydb.cursor()
sql="update patients set address=%s where name=%s"
val=(address,name)
mycursor.execute(sql,val)
mydb.commit()
if(mycursor.rowcount==1):
   print(mycursor.rowcount,"record updated success")
else:
   print(mycursor.rowcount,"record updated failed")
```